

Airbnb Market Analysis

Project Summary

This project looks at trends in the Asheville Airbnb market to better understand pricing, availability, and what makes certain listings perform better than others. Using data from Inside Airbnb, specifically customer reviews and listings, we took a data driven approach to explore both supply and demand. We focused on key metrics like occupancy rate, booking frequency, and estimated monthly revenue to get a sense of how successful different listings are. We also looked at review sentiment to understand what guests actually care about when they stay somewhere. From our early findings, location seems to play a huge role, especially for places near downtown and popular attractions, since they tend to have higher prices and stay booked more often. Property features like the number of bedrooms and amenities also make a big difference in how much revenue a listing can generate. We also started to see that factors like income levels and population density might impact pricing across different areas. Overall, this project helped us better understand what drives success in the Asheville Airbnb market and gave us useful insights that hosts could use to improve their listings.

Analysis of Data and Justification of Methods

Before conducting a deeper analysis, we first performed exploratory data analysis (EDA) to better understand the structure of the datasets. This included examining the distribution of prices, occupancy rates, and the number of reviews, and the typical rating across listings to gauge the overall market. In order for us to establish key performance indicators, we measured market supply by counting active listings and market demand by calculating average occupancy rates. We also created visualizations such as scatter plots, box plots, and bar charts to identify patterns and outliers in the data. For example, we used scatter plots to analyze the relationship between price and occupancy rate, box plots to examine how estimated monthly revenue varies by number of bedrooms, and bar charts to compare listing performance across different neighborhoods. These visualizations helped highlight trends such as the concentration of bookings at lower price levels and differences in revenue across locations, which guided the rest of our analysis.

To further analyze the Airbnb market, we used correlation analysis and feature engineering to better understand how key variables are related. First, we created a correlation matrix using variables such as price, occupancy rate, estimated monthly revenue, and number of bedrooms. This helped us identify the strongest relationships in the data. For example, we found a strong positive relationship between price and estimated monthly revenue, while the relationship between price and occupancy rate was very weak.

We also applied feature engineering to create new variables that better represent listing performance. Specifically, we calculated estimated monthly revenue by multiplying average nightly price by the average number of days booked per month. Additionally, we calculated the occupancy rate by combining the listings data with the calendar datasets so these variables could be analyzed together. All in all, merging these datasets enabled better performance comparisons across listings.

Results/Findings

From the analysis, a few key trends stand out in the Asheville Airbnb market. Pricing does not appear to have a strong relationship with occupancy. Lower-priced listings are generally booked more often, while higher-priced listings sometimes struggle to stay filled. A similar pattern appears with reviews, as listings with more reviews tend to be priced lower, suggesting they are booked more frequently. In contrast, higher-priced listings usually have fewer reviews. Location also plays a big role, with some neighborhoods bringing in much higher monthly revenue than others. Property size matters as well, since listings with more bedrooms typically earn more. However, there are still a few smaller properties that perform surprisingly well. Overall, these results show that success depends on a mix of pricing, location, and property features rather than just one factor.

Recommendations

If you're looking to excel in the Asheville Airbnb scene, there needs to be a tactical approach to setting your nightly rates. There was a high correlation of 0.74 between price and how much revenue you actually bring in; this does not mean that high-priced properties bring in the most revenue. The relationship between price and occupancy rate is super weak at -0.046, which indicates that just because the listing is expensive does not mean people will not book it, but it poses a greater risk. Most of the listings that actually get booked are grouped at lower price points, and once the price climbs, occupancy rate usually takes a dive. Using dynamic pricing is a huge recommendation here so the listing can stay competitive and adjust based on real-time demand.

Another big thing is focusing on your estimated monthly revenue; this was calculated by multiplying the nightly price by the average days you are booked each month. To get those numbers up, there needs to be strategic planning about when your place is available and how to keep guest review scores high. Listings with a ton of reviews usually have lower prices, while expensive properties have fewer reviews, which indicates that the booking frequency of these properties is on the lower end. It is basically a cycle where more reviews lead to increased trust, visibility, and more bookings, so focusing on guest experience and engagement is crucial.

Upgrading the space with better amenities or more room for guests is also a recommended step. Certain outlier properties in the data were small but outperforming bigger properties in sales. That means if a property has a unique appeal or better features, it can perform substantially well compared to other listings. Also, location is everything. A property in the 28805 zip code, has the highest median revenue in the city. On the flip side, areas within the 28732 or 28715 struggle with the lowest revenue numbers. So, if you are thinking about where to buy or how to market your place, checking the neighborhood performance is a must to make sure you are in a high-demand area.